

What is GitOps and why should you care?

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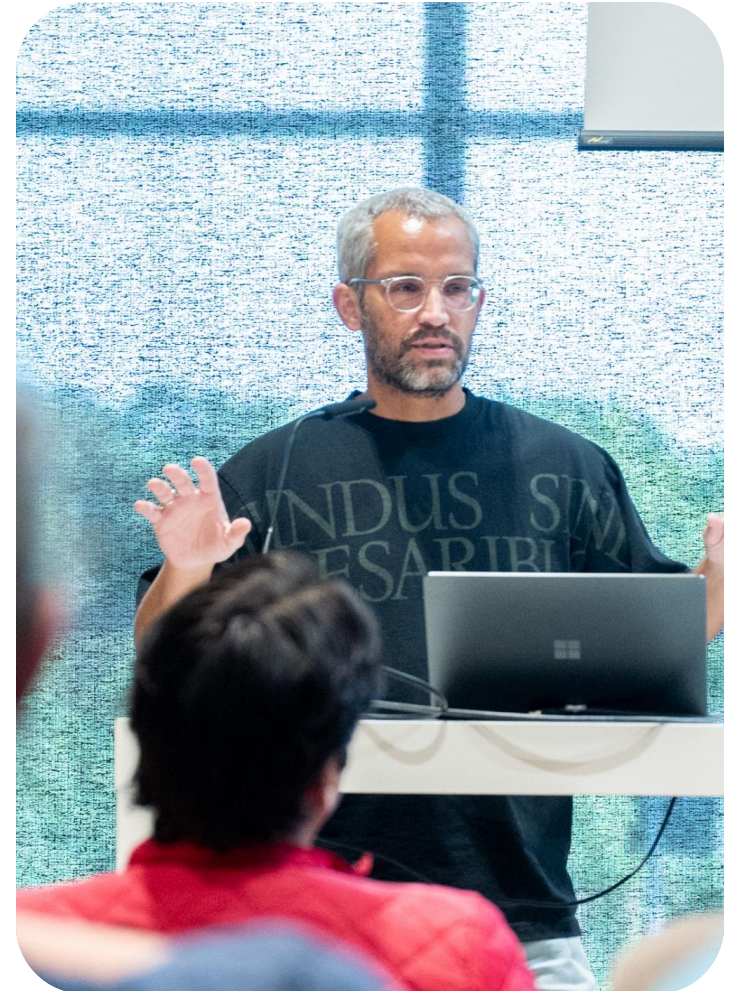
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Agenda

Introduction to the topic and challenge

What is GitOps and why does it help?

Tech stack big picture

Deep dive and live demo

Best practices and lessons learned

Q&A

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Do you still ClickOps? Or already IaC?



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Virtual machine name * ⓘ

Region * ⓘ (Europe) Germany West Central

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Virtual machine

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ml-experimente772-46ce370b

IP configuration ml-experimente772-defaultIpConfigu...

Network Interface: ml-experimente772-46ce370b Effective security rules Troubleshoot VM connection issues Topology

Virtual network/subnet: vnet-germanywestcentral/snet-germanywestcentral-1 NIC Public IP: 4.182.232.69 NIC Private IP: 172.16.0.4
Accelerated networking: Enabled

Inbound port rules Outbound port rules Application security groups Load balancing

Network security group ml-experimente-nsg (attached to network interface: ml-experimente772-46ce370b)
Impacts 0 subnets, 1 network interfaces

Add inbound port rule

Priority	Name	Port	Protocol	Source	Destination	Action
300	⚠ RDP	3389	TCP	Any	Any	✓ Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	✓ Allow

Do you still ClickOps? Or already IaC?

Issues with ClickOps

- Manual work
 - Error-prone
 - Not reproducible reliably
 - Too slow
- No versioning
- No history of who did what, when and why

→ Not a professional approach

Do you still ClickOps? Or already IaC?

```
resource virtualMachines_ml_experimente_name_resource 'Microsoft.Compute/virtualMachines@2024-11-01' = {
  name: virtualMachines_ml_experimente_name
  location: 'germanywestcentral'
  zones: [
    '1'
  ]
  properties: {
    hardwareProfile: {
      vmSize: 'Standard_D8a1ds_v6'
    }
    storageProfile: {
      imageReference: {
        publisher: 'microsoftwindowsdesktop'
        offer: 'windows-11'
        sku: 'win11-24h2-pro'
        version: 'latest'
      }
    }
    osDisk: {
      osType: 'Windows'
      name: '${virtualMachines_ml_experimente_name}_OsDisk_1_13dfbcae423642d49e3dcb9b7f504a9d'
      createOption: 'FromImage'
      caching: 'ReadWrite'
    }
  }
}
```

Do you still ClickOps? Or already IaC?

Issues with Infrastructure as Code (IaC)

- Versioning and history (of the definition) is possible now
- But:
 - Not immune against manual intervention or inadvertent automated changes (“drift”)
 - Not continuously updated (if your IaC tool can even do that), only push-driven
 - Limited traceability

→ A big step in the right direction, but also still big issues

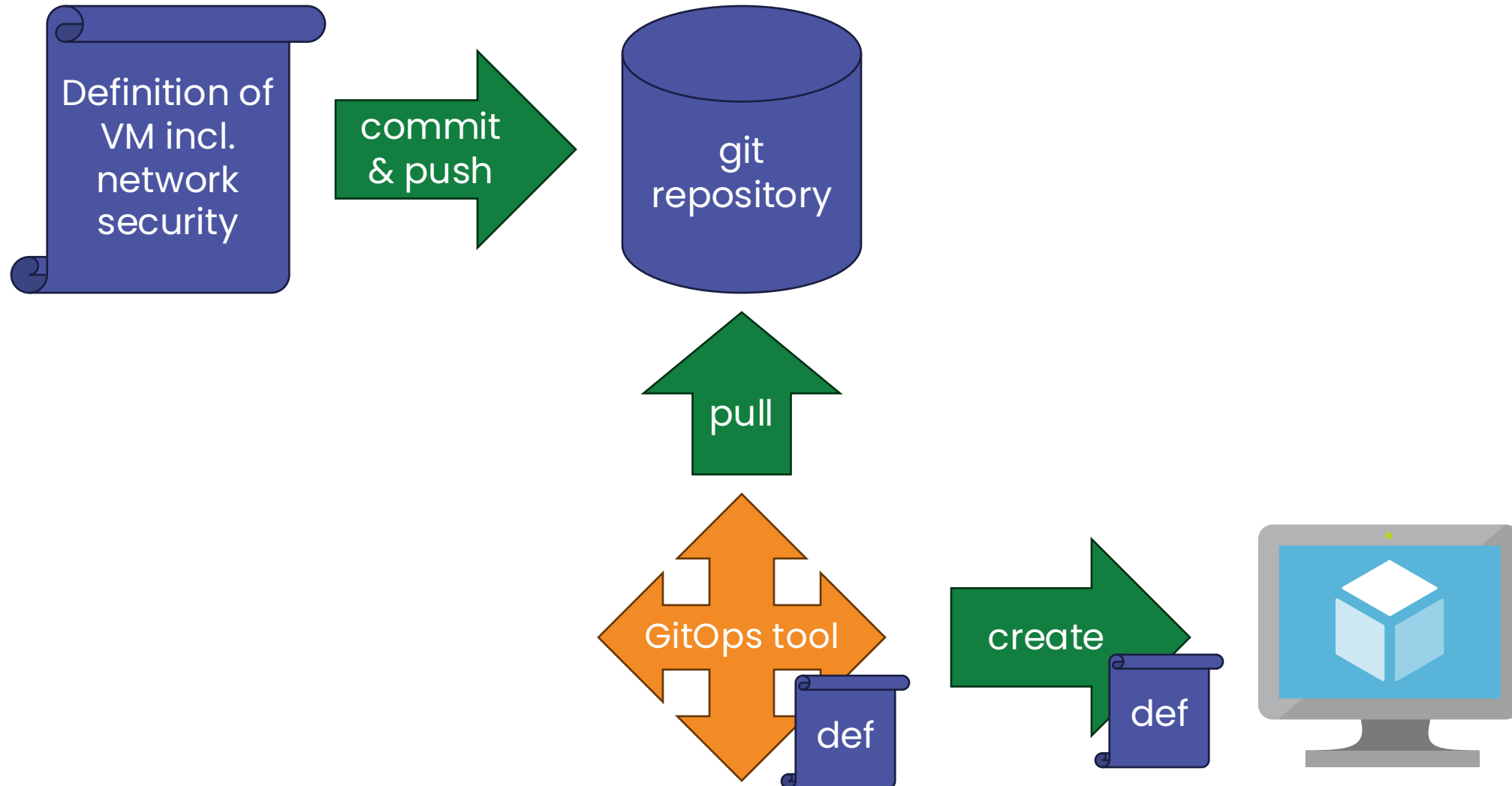
What is GitOps and why does it help?

Base concepts

- Declarative: You describe what you want, not how to get there
 - IaC can also be imperative e.g. with PowerShell or other scripts
- Git as single source of truth
 - No local execution of IaC definitions with dependencies on or influence by anything local
 - Enables security / compliance checks on PR level
- Pull-based
 - IaC typically push-based: "I tell the cloud what to do"
 - GitOps pull-based: A system ensures that reality matches the desired state
- Identifies "drift" and reconciles it
 - Automatically analyses the resource state
 - If required, applies changes to resources so that they match definition
 - Failure to do so can be monitored
- Consistency across environments (e.g. dev / stage / prod)
 - Same code → same setup. Deviations easily identifiable

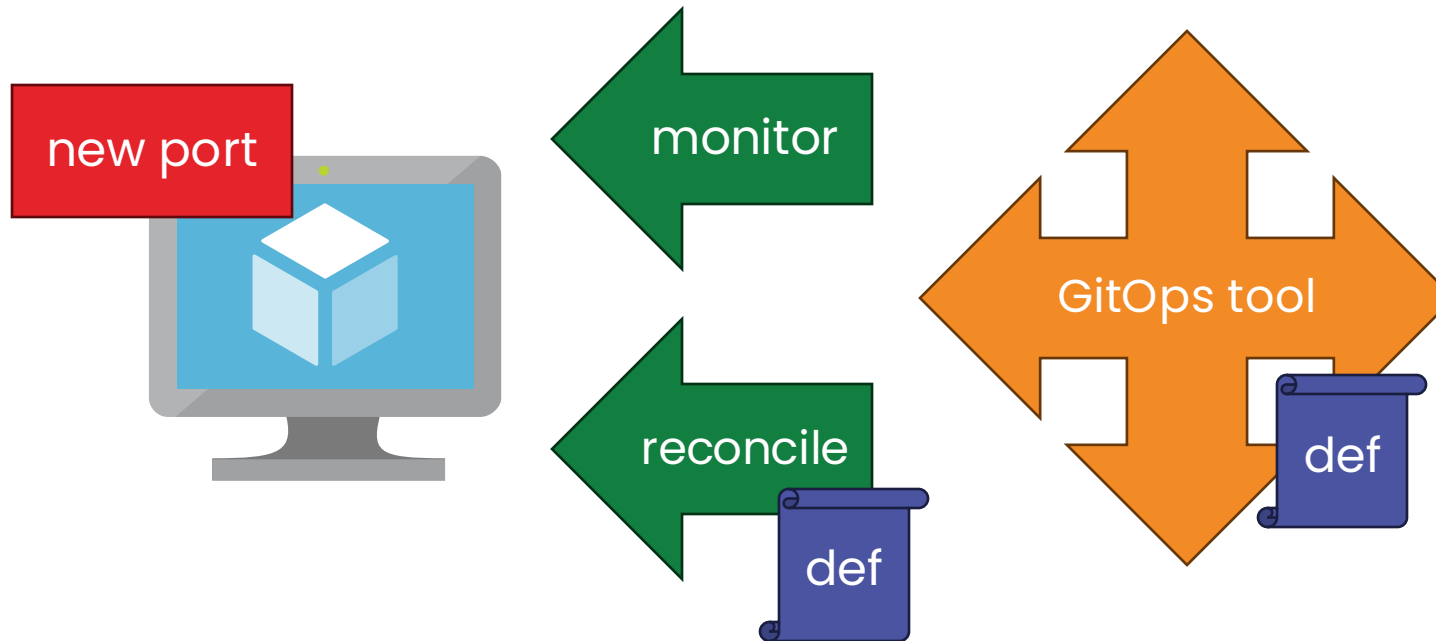
What is GitOps and why does it help?

Base concepts: pull-based with git repository



What is GitOps and why does it help?

Base concepts: drift and reconciliation



Tech stack big picture

- Kubernetes: De-facto standard orchestration engine for container workloads
 - Answer to “I want to run a lot of containers with some dependencies between each other”
 - Invented by Google, used by a lot of the big tech companies (Netflix, Shopify, Paypal, also Microsoft) and trickling down to smaller companies as well as seeping into other scenarios; by now part of CNCF
- Crossplane: Kubernetes-native, GitOps-friendly IaC tool to deploy Cloud resources
 - Answer to “I want to deploy a Kubernetes cluster and dependent resources somewhere in the Cloud”
 - Invented by Upbound, by now part of CNCF
- Argo CD: GitOps tool to deploy applications to Kubernetes
 - Answer to “I want a Business Central container on my Kubernetes cluster”
 - Invented by Applatix, by now part of CNCF



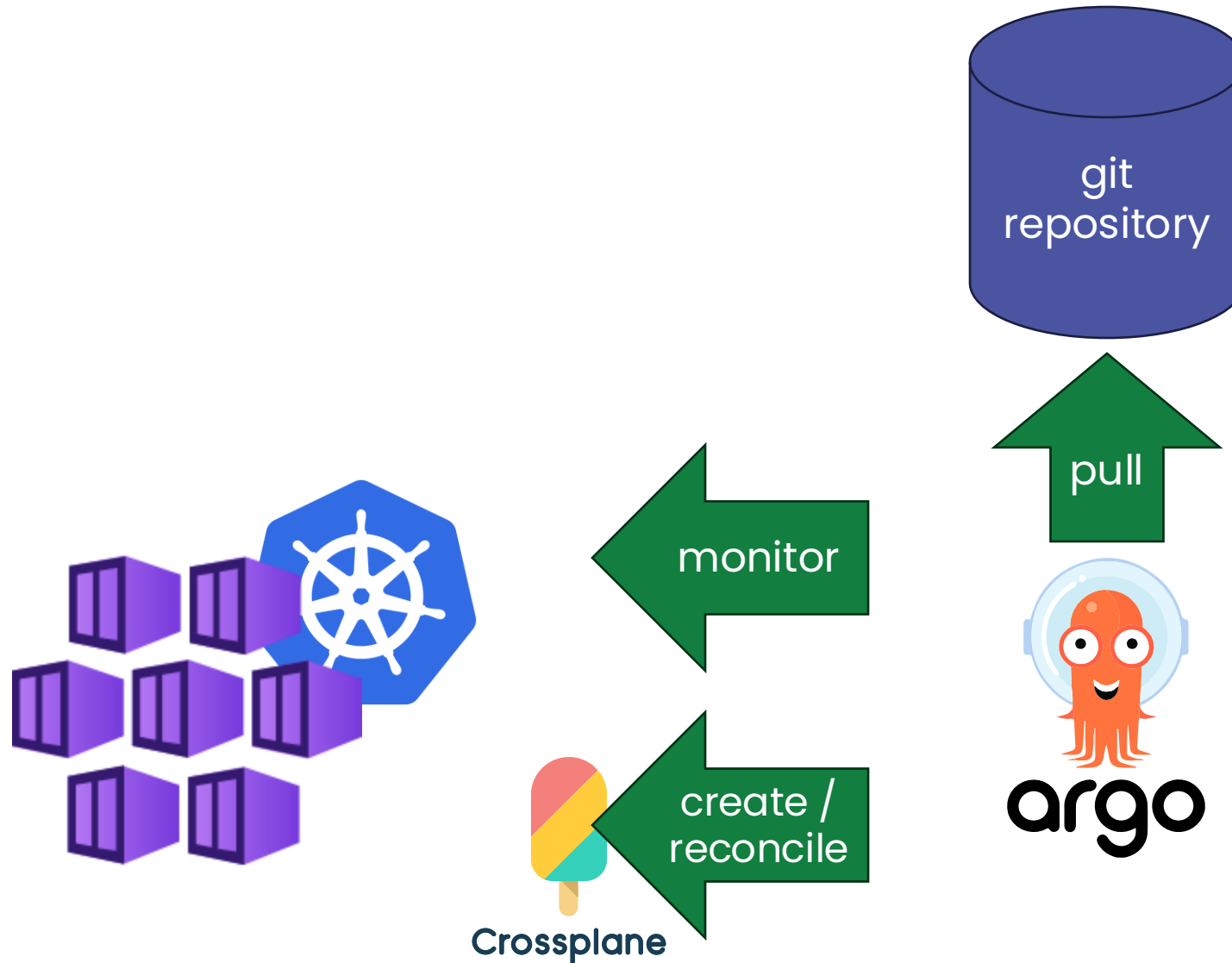
Crossplane



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Tech stack big picture

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Deep dive and live demo

We'll look at

- How to do infrastructure changes the GitOps-way
- Rolling back changes
- Drift detection & drift handling
- Traceability and auditability of changes



Best practices and lessons learned

When to use GitOps

- **GitOps works well in cloud-native scenarios**
 - Managing Kubernetes clusters and its applications, networking, storage, etc.
 - Infrastructure management across multiple cloud platforms from one source
 - Maintaining multiple similar environments (i.e. development, staging & production)
 - Example: one branch for each stage, progressing changes through each stage by simple Pull Requests
- **What to consider before using GitOps**
 - Setting up good GitOps tools, structures, declarative definitions & processes takes time
 - Quick experiments & initial setup might be better done without GitOps first and are integrated later at a stable stage
 - Not everything you might need is available declaratively

Best practices and lessons learned

Wrap up

- GitOps tools like Argo CD: ensure reality matches desired Git state
 - Crossplane integration: secure, consistent, auditable & self-healing infrastructure
 - Git based-workflows: Commit, Pull-Requests & reviews of declarative configs – like with software
-
- ➔ Start small: pilot project, minimal repo structures & simple processes
 - ➔ Grow incrementally
 - ➔ Bonus after Q&A: Real-world GitOps example of COSMO Alpaca

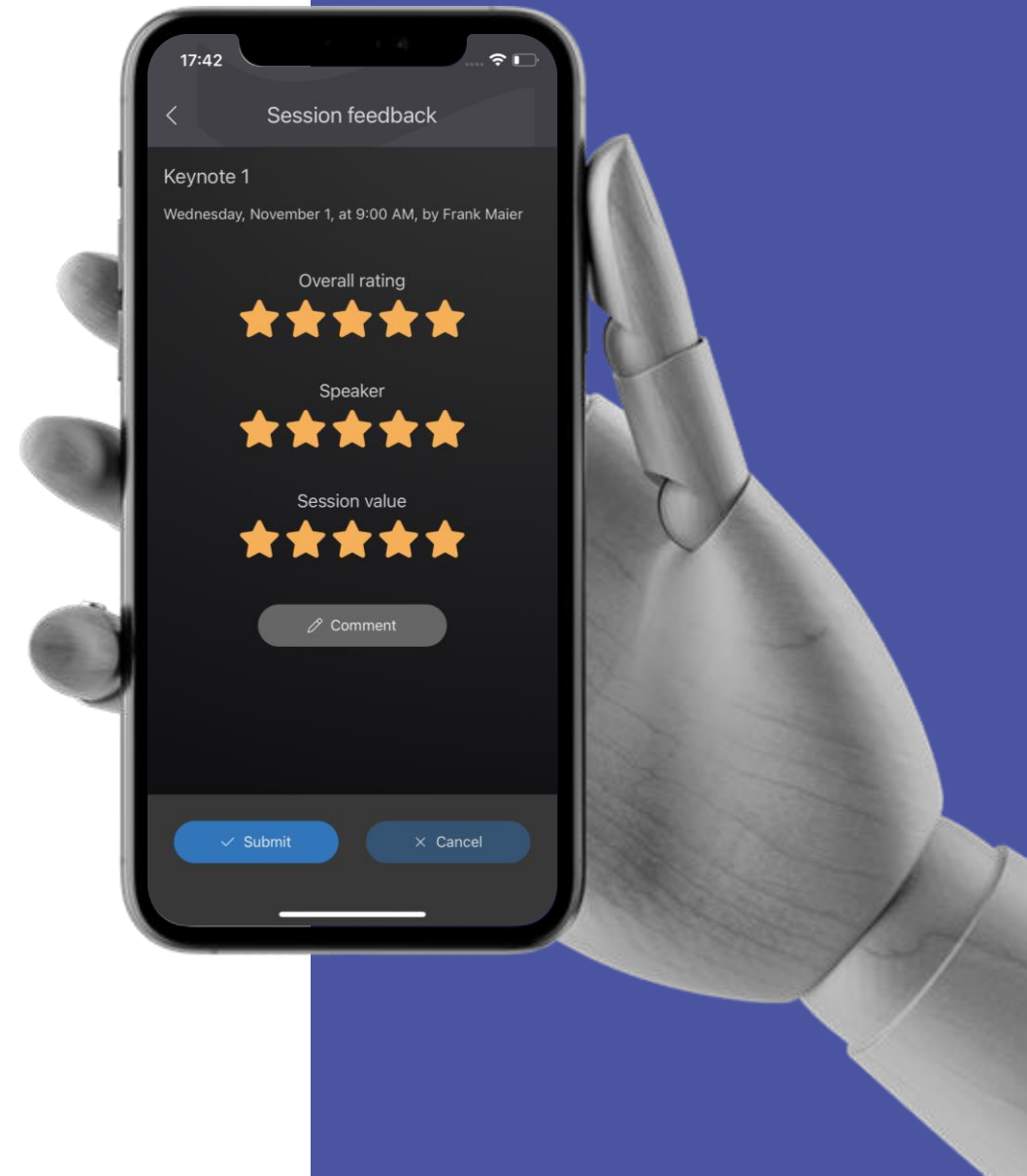
Give us Feedback!

and help us improve!

Please rate our session in the Conference App!

and leave a constructive comment.

Thank you !



Any questions we can try to answer?

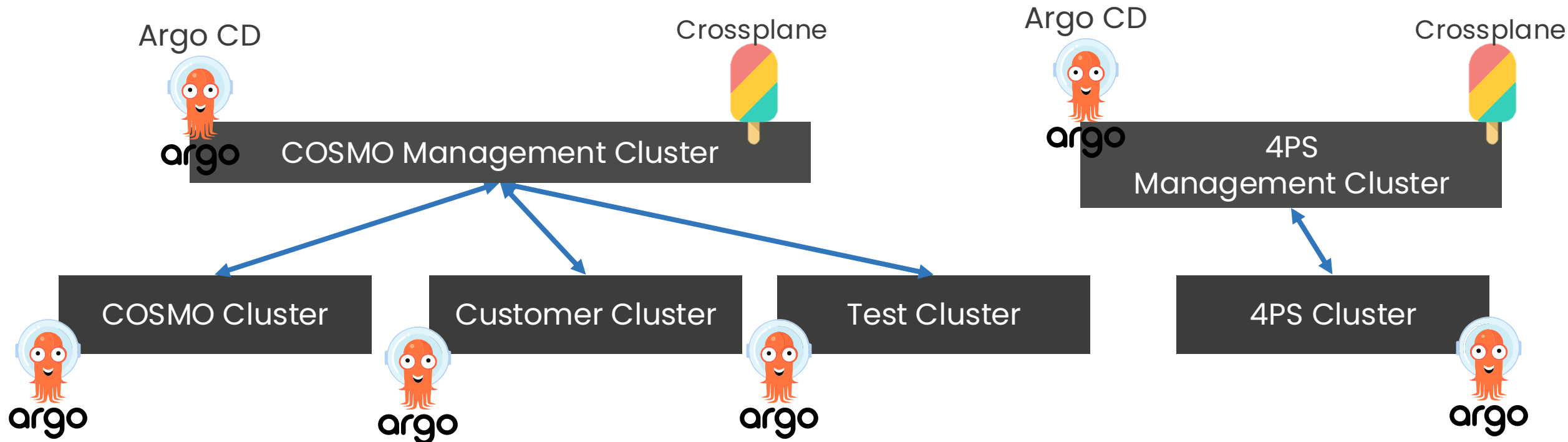


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Bonus: Real World Example

from COSMO Alpaca

- COSMO Alpaca uses Azure Kubernetes Service to host different C# WebAPIs and Business Central containers
- 4 clusters managed via GitOps with Argo CD & Crossplane by COSMO, 2 clusters by 4PS
- Test/Customer/COSMO cluster are similar, progressing changes through those clusters ("stages") via PRs

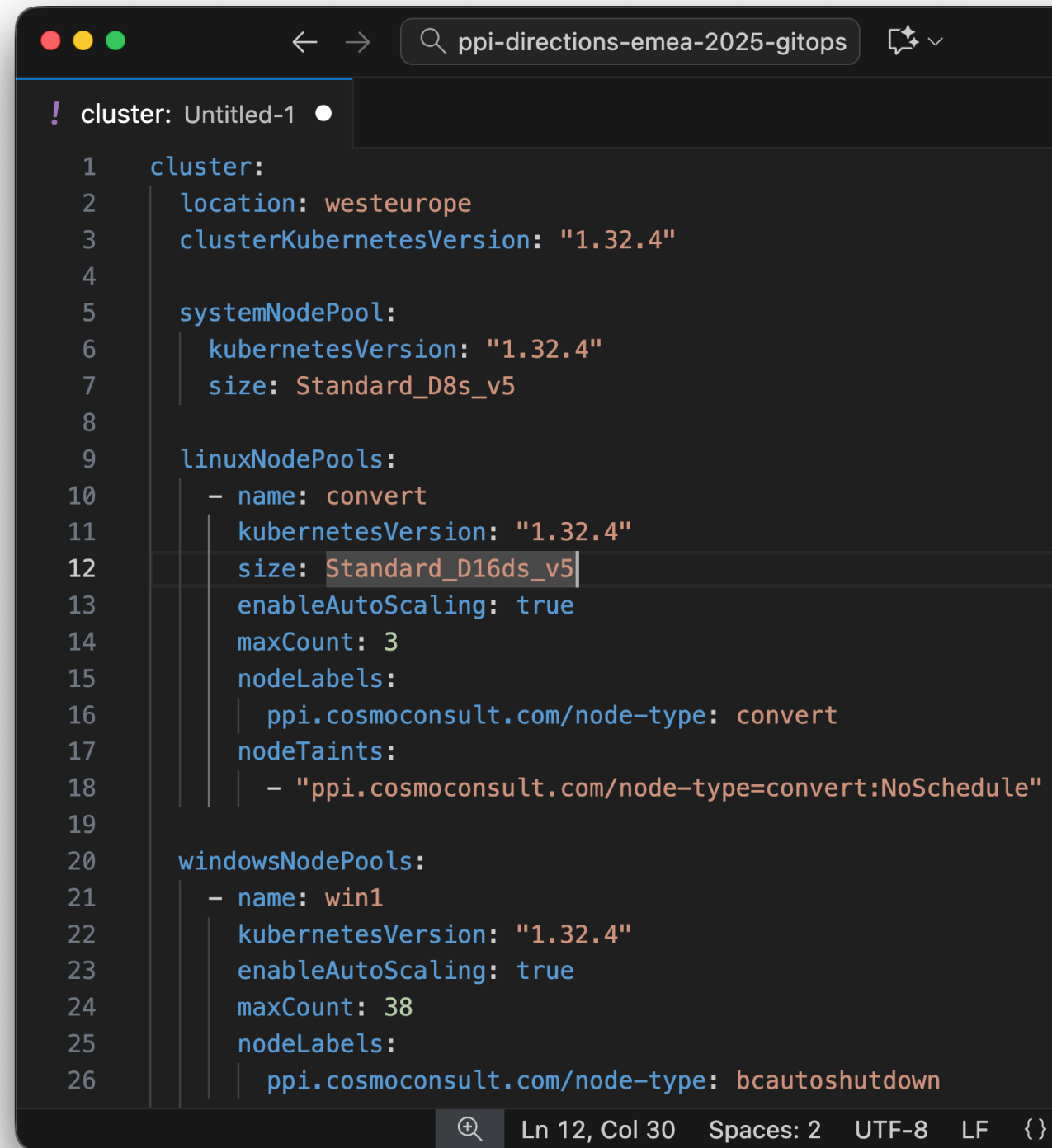


Bonus: Real World Example

from COSMO Alpaca

AKS clusters are defined declaratively:

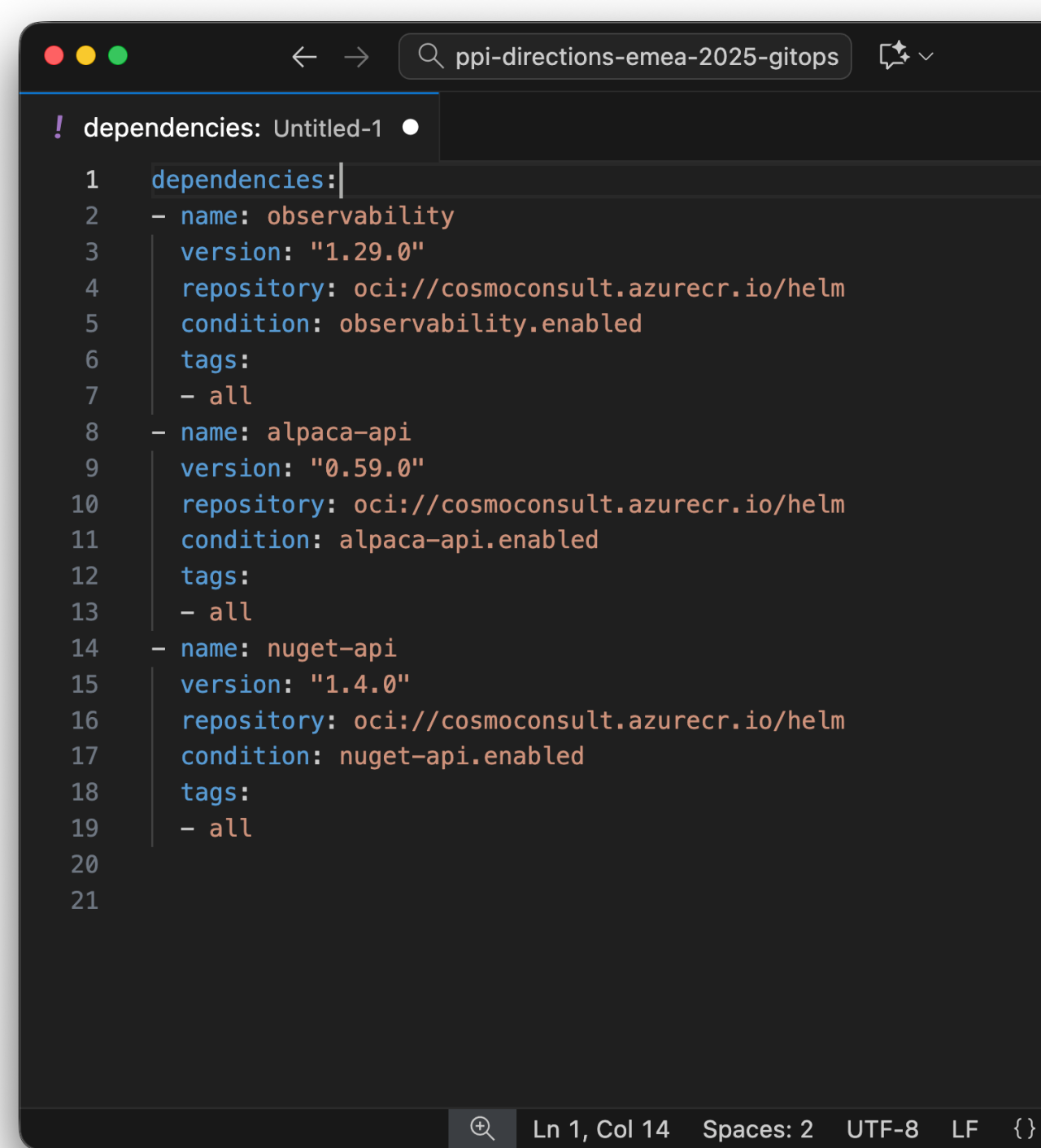
Node size, Node count, Kubernetes version, etc.

A screenshot of a code editor window with a dark theme. The title bar shows three colored window control buttons (red, yellow, green) on the left, and navigation arrows, a search bar containing 'ppi-directions-emea-2025-gitops', and a star icon on the right. The editor content shows a YAML file named 'cluster: Untitled-1'. The code is a Kubernetes cluster definition. Line 1 is 'cluster:'. Line 2 is 'location: westeurope'. Line 3 is 'clusterKubernetesVersion: "1.32.4"'. Line 4 is empty. Line 5 is 'systemNodePool:'. Line 6 is 'kubernetesVersion: "1.32.4"'. Line 7 is 'size: Standard_D8s_v5'. Line 8 is empty. Line 9 is 'linuxNodePools:'. Line 10 is '- name: convert'. Line 11 is 'kubernetesVersion: "1.32.4"'. Line 12 is 'size: Standard_D16ds_v5'. Line 13 is 'enableAutoScaling: true'. Line 14 is 'maxCount: 3'. Line 15 is 'nodeLabels:'. Line 16 is 'ppi.cosmoconsult.com/node-type: convert'. Line 17 is 'nodeTaints:'. Line 18 is '- "ppi.cosmoconsult.com/node-type=convert:NoSchedule"'. Line 19 is empty. Line 20 is 'windowsNodePools:'. Line 21 is '- name: win1'. Line 22 is 'kubernetesVersion: "1.32.4"'. Line 23 is 'enableAutoScaling: true'. Line 24 is 'maxCount: 38'. Line 25 is 'nodeLabels:'. Line 26 is 'ppi.cosmoconsult.com/node-type: bcautoshtutdown'. The status bar at the bottom shows a magnifying glass icon, 'Ln 12, Col 30', 'Spaces: 2', 'UTF-8', 'LF', and a curly brace icon.

Bonus: Real World Example

from COSMO Alpaca

Each cluster has its own Git repository defining which version of each application should be deployed to the respective cluster



The screenshot shows a code editor window with a dark theme. The title bar at the top includes standard macOS window controls (red, yellow, green buttons) and a search bar containing the text "ppi-directions-emea-2025-gitops". The editor displays a file named "dependencies: Untitled-1" with a list of Helm dependencies. The code is as follows:

```
1 dependencies:
2   - name: observability
3     version: "1.29.0"
4     repository: oci://cosmoconsult.azurecr.io/helm
5     condition: observability.enabled
6     tags:
7       - all
8   - name: alpaca-api
9     version: "0.59.0"
10    repository: oci://cosmoconsult.azurecr.io/helm
11    condition: alpaca-api.enabled
12    tags:
13      - all
14   - name: nuget-api
15     version: "1.4.0"
16     repository: oci://cosmoconsult.azurecr.io/helm
17     condition: nuget-api.enabled
18     tags:
19       - all
20
21
```

The status bar at the bottom right indicates the current cursor position is "Ln 1, Col 14", with "Spaces: 2", "UTF-8", "LF", and a brace icon.

Bonus: Real World Example

from COSMO Alpaca

After a new release of our APIs, PRs in the GitOps repositories are automatically created.

Rollout automatically starts after approving a PR.

